

Total No. of Questions : 4]

SEAT No. :

PE-549

[Total No. of Pages : 2

[6578]-22

S.E. (Artificial Intelligence & Data Science)/ (Computer Science & Design Engg.)/ (Computer Engineering) (In Sem.)

COMPUTER GRAPHICS

(2019 Pattern) (Semester - III) (210244)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2 & Q.3 or Q.4.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Draw neat diagram wherever necessary.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Define the following terms: Frame buffer, Resolution, Aspect ratio, Pixel and Display device. [5]
- b) Differentiate between Raster Graphics and Vector Graphics. [5]
- c) How Keyboard and Mouse interaction is done in OpenGL? [5]

OR

- Q2)** a) Derive the expression for decision parameter of Bresenham's Circle Drawing Algorithm. [5]
- b) Explain DDA Line Drawing Algorithm with an example. [5]
- c) Plot 6 points of circle using Bresenham's Circle Drawing Algorithm when radius of circle is 10 units. The circle has centre (50,50). [5]

P.T.O.

- Q3)** a) Define Polygon? Explain Convex and Concave Polygon with figure. [5]
- b) Explain winding number Method of inside test with example. [5]
- c) Write the Psudocode for Cohen-Sutherland line clipping algorithm. [5]

OR

- Q4)** a) Explain Scan line fill algorithm with suitable example. [5]
- b) Write a Psuedocode for Polygon clipping Algorithm. [5]
- c) Explain even-odd method to determine polygon interior points. [5]

